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Lecture 12: Finite Elasticity

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Outline and Motivation

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Key Concept: Finite Elasticity

We will develop an exact theory of elasticity without assuming that deformations are small. This provides a deeper understanding of the underlying physics.

Kinematics: Describing the Motion

We describe motion by mapping particles from a **reference body** to their current position.

- At time t , the body occupies a region $M_t \subset \mathbb{R}^3$.
- We use a fixed reference body, M . Each point $\mathbf{x} \in M$ represents a unique particle.
- The **motion** is a map φ that gives the position of particle $\mathbf{x}$ at time t :

$$\mathbf{y} = \varphi(\mathbf{x}, t)$$

Here, $\mathbf{y} \in M_t$ is the current position. The motion φ is the fundamental field in this theory.

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We assume the motion is smooth and one-to-one (no tearing or fractures).

Key Quantities: Velocity Deformation Gradient

From the motion $\varphi(\mathbf{x}, t)$, we can define several important quantities.

Velocity

The velocity of a particle $\mathbf{x}$ is the time derivative of its trajectory:

$$v_i(\mathbf{x}, t) = \frac{\partial \varphi_i}{\partial t}(\mathbf{x}, t)$$

This is a *material* velocity (for a fixed particle), not a spatial one.

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Deformation Gradient

The deformation gradient, $\mathbf{F}$, quantifies local deformation:

$$F_{ij} = \frac{\partial \varphi_i}{\partial x_j}$$

It describes how the motion differs between nearby points

Key Quantities: The Jacobian

Jacobian (J)

The determinant of $\mathbf{F}$ is the Jacobian. It relates volume elements.

$$J(\mathbf{x}, t) = \det \mathbf{F}(\mathbf{x}, t) > 0$$

$J > 0$ ensures that the orientation of the material is preserved (e.g., a right-handed basis maps to a right-handed basis).

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Transforming Integrals

The Jacobian is crucial for changing variables between the reference and current bodies:

$$\int_{M_t} f(\mathbf{y}) d^3 \mathbf{y} = \int_M f[\varphi(\mathbf{x}, t)] J(\mathbf{x}, t) d^3 \mathbf{x}$$

This proves very useful within the theory.

Conservation of Mass

The principle that mass is conserved for any sub-body leads to a key relationship between densities.

- Let $\rho(\mathbf{y}, t)$ be the density at a point $\mathbf{y}$ in the current configuration M_t .
- Let $\rho_0(\mathbf{x})$ be a time-independent density defined in the reference body M .

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Mass Conservation Equation

The relationship between the current (*spatial*) density and the *referential* density is:

$$\rho_0(\mathbf{x}) = J(\mathbf{x}, t)\rho[\varphi(\mathbf{x}, t), t]$$

This is a statement of mass conservation expressed in the reference body.

Energy of an Elastic Body

Kinetic Energy ($\mathcal{T}$)

The total kinetic energy is the integral of the kinetic energy density over the reference body:

$$\mathcal{T}(t) = \frac{1}{2} \int_M \rho_0(\mathbf{x}) \|\mathbf{v}(\mathbf{x}, t)\|^2 d^3\mathbf{x}$$

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Potential Energy ($\mathcal{V}$)

For a **simple hyperelastic body**, potential energy is stored due to deformation and depends locally on the deformation gradient $\mathbf{F}$:

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \mathbf{F}(\mathbf{x}, t)] d^3\mathbf{x}$$

The function W is the **strain energy function**.

Symmetries of Potential Energy (1/2)

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Symmetry 1: Time-Translation Invariance

The material's properties do not change over time, so physics should be the same regardless of when you start the clock.

- This requires that the potential energy density W has no explicit dependence on time, t .

Symmetries of Potential Energy (2/2)

Symmetry 2: Invariance Under Rigid Body Motions

The potential energy should not change if the body is simply translated or rotated without changing its shape or volume.

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The potential energy should not change if the body is simply translated or rotated without changing its shape or volume.

- Invariance under translation implies W cannot depend on the motion φ itself, only on the deformation gradient $\mathbf{F}$.
- Invariance under rotation requires that for any rotation matrix $\mathbf{Q}$:

$$W(\mathbf{x}, \mathbf{F}) = W(\mathbf{x}, \mathbf{QF})$$

The Principle of Stationary Action

The equations of motion can be derived from a variational principle.

Lagrangian Density ($\mathcal{L}$)

The Lagrangian density is the kinetic minus the potential energy density:

$$\mathcal{L} = \frac{1}{2}\rho_0(\mathbf{x})\|\mathbf{v}\|^2 - W(\mathbf{x}, \mathbf{F})$$

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Action ($\mathcal{S}$)

The action is the integral of the Lagrangian density over the reference body and a time interval $[0, T]$:

$$\mathcal{S}(\varphi) = \int_0^T \int_M \mathcal{L} d^3\mathbf{x} dt$$

Hamilton's Principle

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The true motion $\varphi(\mathbf{x}, t)$ is a stationary point of the action, $\delta\mathcal{S} = 0$, for all smooth variations $\delta\varphi$ that vanish at the start and end times, $t = 0$ and $t = T$.

This powerful principle turns a dynamics problem into a calculus of variations problem.

Deriving the Equations of Motion

Applying Hamilton's principle ($\delta\mathcal{S} = 0$) yields the Euler-Lagrange equations.

Euler-Lagrange Equations (General Form)

For a general Lagrangian density $\mathcal{L}(\varphi, \mathbf{v}, \mathbf{F})$:

$$\frac{\partial \mathcal{L}}{\partial \varphi_i} - \frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \right) - \frac{\partial}{\partial x_j} \left(\frac{\partial \mathcal{L}}{\partial F_{ij}} \right) = 0$$

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First Piola-Kirchhoff Stress Tensor

We define the stress tensor as the derivative of the strain energy with respect to the deformation gradient:

$$T_{ij} = \frac{\partial W}{\partial F_{ij}}$$

It relates forces in the current body to areas in the reference

The Final Equation of Motion

Substituting the specific Lagrangian for an elastic body into the Euler-Lagrange equations gives the final result.

- The time derivative term becomes the rate of change of momentum: $\frac{\partial}{\partial t} \left(\frac{\partial \mathcal{L}}{\partial v_i} \right) = \rho_0 \frac{\partial v_i}{\partial t}.$
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Equation of Motion for an Elastic Body

Putting it all together, we get:

$$\rho_0 \frac{\partial v_i}{\partial t} - \frac{\partial T_{ij}}{\partial x_j} = 0$$

Boundary Conditions

Hamilton's principle also provides the **natural boundary conditions** that must hold on the surface of the body, ∂M .

Natural Boundary Condition

The variation $\delta\mathcal{S}$ is zero only if the following holds on the boundary:

$$T_{ij}\hat{n}_j = 0$$

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- The vector $\mathbf{t} = \mathbf{T}\hat{\mathbf{n}}$ is the **traction**, or surface force per unit reference area.
- This condition means the body has a **traction-free** surface.

Summary: What You Need to Know

- **Kinematics:** Understand the concepts of the reference body, current configuration, motion (φ), deformation gradient ($\mathbf{F}$), and Jacobian (J).

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- **Mass Conservation:** Be able to derive and use the relation between the spatial density ρ and the referential density ρ_0 .
- **Energy:** State the expressions for the kinetic and potential energy of a simple hyperelastic body and explain the underlying assumptions.
- **Derivation:** Use Hamilton's principle to derive the equations of motion and associated boundary conditions for finite elasticity.